Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: CDF Hilmar (Formerly River Rock Ranches LLC)

Physical address of dairy:

18128 American AVE	Hilmar	Merced	95324
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1956

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X160-X065-XXXX

B. OPERATORS

California Dairy Farms

Operator name: California Dairy Farms

Telephone no.: (209) 285-0001 (Landline) (Cellular)

6071 Larson AVE	Hilmar	CA	95324
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Forebay Farms LLC, /Fagundes, Fagundes, Fagundes GP

Legal owner name: Forebay Farms LLC, /Fagundes, Fagundes, Fagundes GP

Telephone no.: (209) 383-6046 (Landline) (Cellular)

P.O. Box 2717	Merced	CA	95344
Mailing Address Number and Street	City	State	Zip Code

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	212	200	300	0	0
Number under roof	1,200	0	0	0	0	0
Maximum number	1,280	220	440	400	150	0
Average number	1,200	212	200	300	0	0
Avg live weight (lbs)	1,000	1,000	750	500		

Predominant milk cow breed: Jersey

Average milk production: 60 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 36,649.05 *tons per reporting period*

Total nitrogen from manure: 470,182.60 *lbs per reporting period*

After ammonia losses (30% loss applied): 329,127.82 *lbs per reporting period*

Total phosphorus from manure: 77,884.74 *lbs per reporting period*

Total potassium from manure: 221,927.07 *lbs per reporting period*

Total salt from manure: 613,769.40 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 18,914,400 *gallons*

Total nitrogen generated: 35,054.93 *lbs*

Total phosphorus generated: 9,414.72 *lbs*

Total potassium generated: 73,087.59 *lbs*

Total salt generated: 388,477.31 *lbs*

	18,914,400	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	18,914,400	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
10	14	14	2	process wastewater	X045-X160-X072-XXXX
11	17	17	2	process wastewater	X045-X160-X072-XXXX
15	3	3	2	process wastewater	X045-X160-X055-XXXX
18	25	25	2	process wastewater	X045-X160-X065-X000
19	8	8	2	process wastewater	X045-X160-X066-XXXX
5	10	10	2	process wastewater	X045-X160-X045-XXXX
Almonds 1	422	422	0	none	X046-X190-X007-XXXX X046-X190-X029-XXXX X046-X190-X031-XXXX X046-X190-X032-XXXX X046-X190-X045-XXXX X046-X190-X049-XXXX
Totals for areas that were used for application	77	77	12		
Totals for areas that were not used for application	422	422	0		
Land application area totals	499	499	12		

B. CROPS AND HARVESTS

10

Field name: 10

11/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 14 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	257.00 *ton*	As-is		67.6	5,000.00	1,100.00	8,800.00		8.15

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.36	183.57	40.39	323.09	969.48

10

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 14 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	403.00 *ton*	As-is		68.8	3,700.00	800.00	4,800.00		5.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.79	213.01	46.06	276.34	987.93

11

Field name: 11

11/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 17 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	172.00 *ton*	As-is		68.6	6,900.00	1,100.00	8,800.00		13.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.12	139.62	22.26	178.07	845.07

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	384.00 *ton*	As-is		70.8	4,400.00	1,000.00	5,400.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	22.59	198.78	45.18	243.95	831.07

15

Field name: 15

11/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 3 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	48.00 *ton*	As-is		71.4	5,100.00	1,300.00	9,100.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.00	163.20	41.60	291.20	951.81

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 3 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	66.00 *ton*	As-is		70.6	4,200.00	1,400.00	6,800.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	22.00	184.80	61.60	299.20	802.03

18

Field name: 18

11/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 25 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	393.00 *ton*	As-is		75.8	4,100.00	1,100.00	8,500.00		18.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.72	128.90	34.58	267.24	1,438.00

18

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	645.00 *ton*	As-is		69.4	3,900.00	800.00	5,100.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.80	201.24	41.28	263.16	821.06

19

Field name: 19

11/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 8 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	90.00 *ton*	As-is		71.6	5,500.00	1,000.00	8,400.00		9.09

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.25	123.75	22.50	189.00	580.85

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 8 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	206.00 *ton*	As-is		72.0	3,700.00	1,000.00	5,400.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.75	190.55	51.50	278.10	951.72

5

Field name: 5

11/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 10 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	111.00 *ton*	As-is		68.6	6,900.00	1,100.00	8,800.00		13.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.10	153.18	24.42	195.36	927.12

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 10 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	270.00 *ton*	As-is		68.2	3,700.00	800.00	4,900.00		4.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.00	199.80	43.20	264.60	772.74

NUTRIENT BUDGET

A. LAND APPLICATIONS

10 - 11/10/2022: Oats, silage-soft dough

Field name: 10

Crop: Oats, silage-soft dough

Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	70.74	13.23	161.25	856.20	342,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	326.72	2,549,400.00 *gal*
Application event totals		70.74	13.23	161.25	1,182.92	

01/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.04	6.26	100.44	507.50	198,000.00 *gal*
Application event totals		31.04	6.26	100.44	507.50	

02/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.51	7.96	127.83	645.90	252,000.00 *gal*
Application event totals		39.51	7.96	127.83	645.90	

03/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	22.57	4.55	73.04	369.09	144,000.00 *gal*
Application event totals		22.57	4.55	73.04	369.09	

04/20/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.33	10.86	100.64	648.05	360,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	231.29	1,804,740.00 *gal*
Application event totals		31.33	10.86	100.64	879.34	

10 - 06/08/2023: Corn, silage

Field name: 10

Crop: Corn, silage

Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.99	16.29	150.96	972.07	540,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	256.33	2,000,128.00 *gal*
Application event totals		46.99	16.29	150.96	1,228.40	

06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	234.12	1,826,808.00 *gal*
Application event totals		0.00	0.00	0.00	234.12	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	245.73	1,917,423.00 *gal*
Application event totals		0.00	0.00	0.00	245.73	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.42	13.99	57.51	270.38	360,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	219.01	1,708,954.00 *gal*
Application event totals		47.42	13.99	57.51	489.39	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.31	11.89	48.88	229.82	306,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	167.23	1,304,892.00 *gal*
Application event totals		40.31	11.89	48.88	397.05	

10 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	61.65	18.19	74.76	351.49	468,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	185.21	1,445,210.00 *gal*
Application event totals		61.65	18.19	74.76	536.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.42	13.99	57.51	270.38	360,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	200.98	1,568,280.00 *gal*
Application event totals		47.42	13.99	57.51	471.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	167.64	1,308,116.00 *gal*
Application event totals		0.00	0.00	0.00	167.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	37.94	11.19	46.01	216.30	288,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	159.25	1,242,613.00 *gal*
Application event totals		37.94	11.19	46.01	375.55	

11 - 11/10/2022: Oats, silage-soft dough

Field name: 11

Crop: Oats, silage-soft dough

Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

11 - 11/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	79.72	14.91	181.72	964.88	468,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	308.70	2,924,986.00 *gal*
Application event totals		79.72	14.91	181.72	1,273.58	

12/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	27.89	5.62	90.23	455.93	216,000.00 *gal*
Application event totals		27.89	5.62	90.23	455.93	

01/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	32.53	6.56	105.27	531.92	252,000.00 *gal*
Application event totals		32.53	6.56	105.27	531.92	

02/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	18.59	3.75	60.15	303.95	144,000.00 *gal*
Application event totals		18.59	3.75	60.15	303.95	

04/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.80	8.94	82.88	533.69	360,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	236.85	2,244,153.00 *gal*
Application event totals		25.80	8.94	82.88	770.53	

11 - 06/08/2023: Corn, silage

Field name: 11

Crop: Corn, silage

Plant date: 06/08/2023

11 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.28	14.31	132.61	853.90	576,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	268.92	2,548,079.00 *gal*
Application event totals		41.28	14.31	132.61	1,122.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	248.72	2,356,642.00 *gal*
Application event totals		0.00	0.00	0.00	248.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	233.28	2,210,340.00 *gal*
Application event totals		0.00	0.00	0.00	233.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.82	14.40	59.20	278.33	450,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	224.29	2,125,136.00 *gal*
Application event totals		48.82	14.40	59.20	502.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.05	11.52	47.36	222.66	360,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	215.58	2,042,652.00 *gal*
Application event totals		39.05	11.52	47.36	438.25	

11 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.77	14.98	61.57	289.46	468,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	208.22	1,972,935.00 *gal*
Application event totals		50.77	14.98	61.57	497.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.68	16.13	66.30	311.73	504,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	194.44	1,842,335.00 *gal*
Application event totals		54.68	16.13	66.30	506.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	169.51	1,606,117.00 *gal*
Application event totals		0.00	0.00	0.00	169.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.01	12.10	49.73	233.80	378,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	157.17	1,489,216.00 *gal*
Application event totals		41.01	12.10	49.73	390.97	

15 - 11/10/2022: Oats, silage-soft dough

Field name: 15

Crop: Oats, silage-soft dough

Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

15 - 11/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.75	6.50	79.21	420.59	36,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	304.83	509,694.00 *gal*
Application event totals		34.75	6.50	79.21	725.41	

12/29/2022	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.51	7.96	127.83	645.90	54,000.00 *gal*
Application event totals		39.51	7.96	127.83	645.90	

01/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.67	10.61	170.44	861.20	72,000.00 *gal*
Application event totals		52.67	10.61	170.44	861.20	

02/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.51	7.96	127.83	645.90	54,000.00 *gal*
Application event totals		39.51	7.96	127.83	645.90	

04/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	21.93	7.60	70.45	453.63	54,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	251.01	419,700.00 *gal*
Application event totals		21.93	7.60	70.45	704.64	

15 - 06/08/2023: Corn, silage

Field name: 15

Crop: Corn, silage

Plant date: 06/08/2023

15 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.79	22.80	211.35	1,360.90	162,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	254.56	425,645.00 *gal*
Application event totals		65.79	22.80	211.35	1,615.46	

06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	226.87	379,348.00 *gal*
Application event totals		0.00	0.00	0.00	226.87	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	240.63	402,344.00 *gal*
Application event totals		0.00	0.00	0.00	240.63	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.20	9.79	40.26	189.26	54,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	226.65	378,975.00 *gal*
Application event totals		33.20	9.79	40.26	415.91	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.26	13.06	53.68	252.35	72,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	215.84	360,909.00 *gal*
Application event totals		44.26	13.06	53.68	468.20	

15 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.26	13.06	53.68	252.35	72,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	209.84	350,877.00 *gal*
Application event totals		44.26	13.06	53.68	462.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	11.07	3.26	13.42	63.09	18,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	192.68	322,177.00 *gal*
Application event totals		11.07	3.26	13.42	255.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	172.28	288,070.00 *gal*
Application event totals		0.00	0.00	0.00	172.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.39	19.59	80.51	378.53	108,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	156.52	261,706.00 *gal*
Application event totals		66.39	19.59	80.51	535.04	

18 - 11/10/2022: Oats, silage-soft dough

Field name: 18

Crop: Oats, silage-soft dough

Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

18 - 11/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/27/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.21	10.14	123.57	656.12	468,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	301.81	4,205,375.00 *gal*
Application event totals		54.21	10.14	123.57	957.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	23.70	4.78	76.70	387.54	270,000.00 *gal*
Application event totals		23.70	4.78	76.70	387.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.28	5.10	81.81	413.38	288,000.00 *gal*
Application event totals		25.28	5.10	81.81	413.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.41	9.55	153.39	775.08	540,000.00 *gal*
Application event totals		47.41	9.55	153.39	775.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	21.93	7.60	70.45	453.63	450,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	235.49	3,281,375.00 *gal*
Application event totals		21.93	7.60	70.45	689.13	

18 - 06/08/2023: Corn, silage

Field name: 18

Crop: Corn, silage

Plant date: 06/08/2023

18 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.86	15.20	140.90	907.27	900,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	257.32	3,585,500.00 *gal*
Application event totals		43.86	15.20	140.90	1,164.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	212.28	2,957,900.00 *gal*
Application event totals		0.00	0.00	0.00	212.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	216.93	3,022,725.00 *gal*
Application event totals		0.00	0.00	0.00	216.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.11	15.67	64.41	302.82	720,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	223.93	3,120,200.00 *gal*
Application event totals		53.11	15.67	64.41	526.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	63.74	18.80	77.29	363.39	864,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	215.89	3,008,150.00 *gal*
Application event totals		63.74	18.80	77.29	579.27	

18 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.84	11.75	48.31	227.12	540,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	206.78	2,881,250.00 *gal*
Application event totals		39.84	11.75	48.31	433.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.49	12.54	51.53	242.26	576,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	195.05	2,717,875.00 *gal*
Application event totals		42.49	12.54	51.53	437.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	174.01	2,424,613.00 *gal*
Application event totals		0.00	0.00	0.00	174.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.20	9.79	40.26	189.26	450,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	159.09	2,216,708.00 *gal*
Application event totals		33.20	9.79	40.26	348.35	

19 - 11/10/2022: Oats, silage-soft dough

Field name: 19

Crop: Oats, silage-soft dough

Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

19 - 11/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/27/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.15	12.19	148.52	788.60	180,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	305.93	1,364,096.00 *gal*
Application event totals		65.15	12.19	148.52	1,094.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.51	7.96	127.83	645.90	144,000.00 *gal*
Application event totals		39.51	7.96	127.83	645.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/28/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	24.69	4.98	79.89	403.69	90,000.00 *gal*
Application event totals		24.69	4.98	79.89	403.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	19.75	3.98	63.91	322.95	72,000.00 *gal*
Application event totals		19.75	3.98	63.91	322.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	19.19	6.65	61.64	396.93	126,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	245.95	1,096,640.00 *gal*
Application event totals		19.19	6.65	61.64	642.88	

19 - 06/08/2023: Corn, silage

Field name: 19

Crop: Corn, silage

Plant date: 06/08/2023

19 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.83	19.00	176.12	1,134.09	360,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	280.48	1,250,608.00 *gal*
Application event totals		54.83	19.00	176.12	1,414.56	

06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	264.85	1,180,920.00 *gal*
Application event totals		0.00	0.00	0.00	264.85	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	235.21	1,048,784.00 *gal*
Application event totals		0.00	0.00	0.00	235.21	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.52	10.77	44.28	208.19	158,400.00 *gal*
Canal	Surface water	0.00	0.00	0.00	220.86	984,792.00 *gal*
Application event totals		36.52	10.77	44.28	429.05	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.20	9.79	40.26	189.26	144,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	192.16	856,816.00 *gal*
Application event totals		33.20	9.79	40.26	381.42	

19 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.79	14.69	60.38	283.90	216,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	161.58	720,464.00 *gal*
Application event totals		49.79	14.69	60.38	445.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.50	12.24	50.32	236.58	180,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	181.30	808,413.00 *gal*
Application event totals		41.50	12.24	50.32	417.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	154.33	688,136.00 *gal*
Application event totals		0.00	0.00	0.00	154.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.65	13.47	55.35	260.24	198,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	146.33	652,460.00 *gal*
Application event totals		45.65	13.47	55.35	406.57	

5 - 11/10/2022: Oats, silage-soft dough

Field name: 5

Crop: Oats, silage-soft dough

Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

5 - 11/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.12	9.75	118.82	630.88	180,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	307.04	1,711,326.00 *gal*
Application event totals		52.12	9.75	118.82	937.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.41	9.55	153.39	775.08	216,000.00 *gal*
Application event totals		47.41	9.55	153.39	775.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.46	8.76	140.61	710.49	198,000.00 *gal*
Application event totals		43.46	8.76	140.61	710.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/13/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	23.70	4.78	76.70	387.54	108,000.00 *gal*
Application event totals		23.70	4.78	76.70	387.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	17.54	6.08	56.36	362.91	144,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	249.23	1,389,116.00 *gal*
Application event totals		17.54	6.08	56.36	612.14	

5 - 06/08/2023: Corn, silage

Field name: 5

Crop: Corn, silage

Plant date: 06/08/2023

5 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.86	15.20	140.90	907.27	360,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	271.44	1,512,880.00 *gal*
Application event totals		43.86	15.20	140.90	1,178.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	262.27	1,461,770.00 *gal*
Application event totals		0.00	0.00	0.00	262.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	236.64	1,318,913.00 *gal*
Application event totals		0.00	0.00	0.00	236.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.16	12.73	52.33	246.04	234,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	222.48	1,240,010.00 *gal*
Application event totals		43.16	12.73	52.33	468.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.84	11.75	48.31	227.12	216,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	220.14	1,226,960.00 *gal*
Application event totals		39.84	11.75	48.31	447.26	

5 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.11	15.67	64.41	302.82	288,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	159.82	890,776.00 *gal*
Application event totals		53.11	15.67	64.41	462.64	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.79	14.69	60.38	283.90	270,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	171.19	954,167.00 *gal*
Application event totals		49.79	14.69	60.38	455.09	

08/10/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	194.53	1,084,213.00 *gal*
Application event totals		0.00	0.00	0.00	194.53	

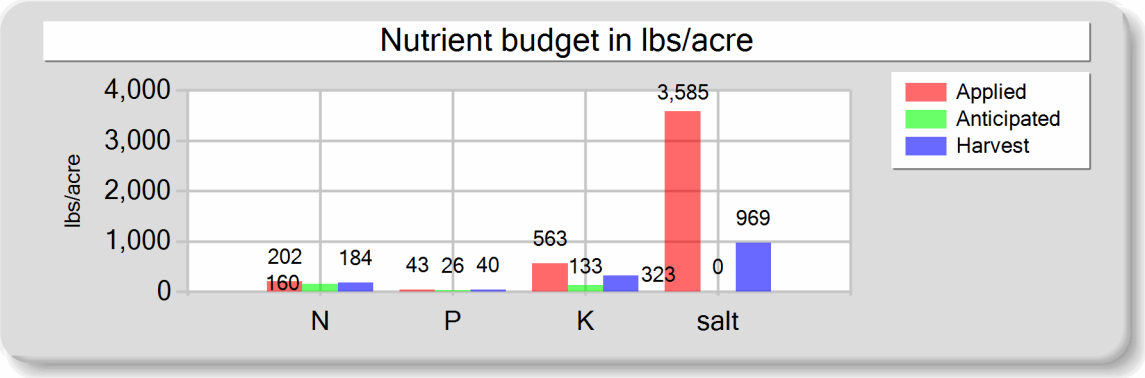
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.84	11.75	48.31	227.12	216,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	179.80	1,002,116.00 *gal*
Application event totals		39.84	11.75	48.31	406.91	

B. NUTRIENT BUDGET

10 - 11/10/2022: Oats, silage-soft dough

Field name: 10 Crop: Oats, silage-soft dough Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	195.19	42.85	563.20	3,026.73
Fresh water	0.00	0.00	0.00	558.01
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	202.19	42.85	563.20	3,584.74
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	183.57	40.39	323.09	969.48
Nutrient balance	18.62	2.47	240.12	2,615.26
Applied to removed ratio	1.10	1.06	1.74	3.70

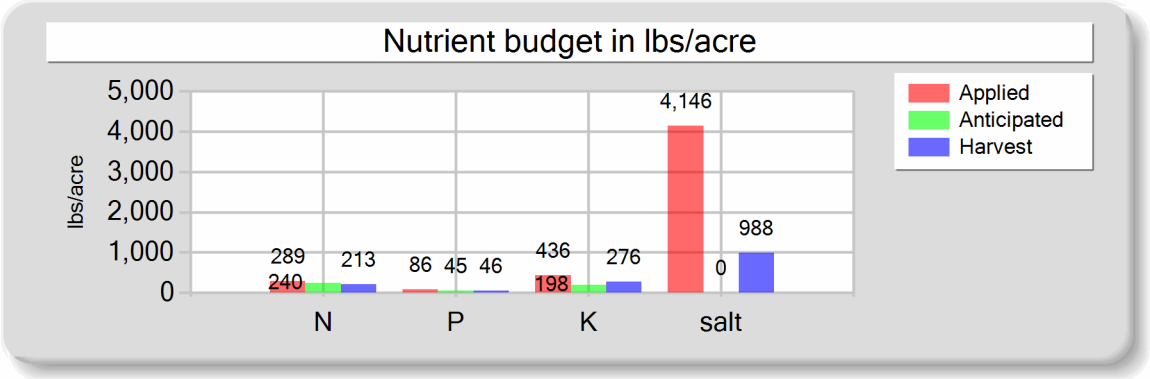
Fresh water applied	
4,354,140.00	*gallons*
160.35	*acre-inches*
11.45	*inches/acre*

Process wastewater applied	
1,296,000.00	*gallons*
47.73	*acre-inches*
3.41	*inches/acre*

Total harvests for the crop	
1	*harvests*

10 - 06/08/2023: Corn, silage

Field name: 10 Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	281.74	85.54	435.63	2,310.44
Fresh water	0.00	0.00	0.00	1,835.50
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	288.74	85.54	435.63	4,145.94
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	213.01	46.06	276.34	987.93
Nutrient balance	75.73	39.49	159.29	3,158.01
Applied to removed ratio	1.36	1.86	1.58	4.20

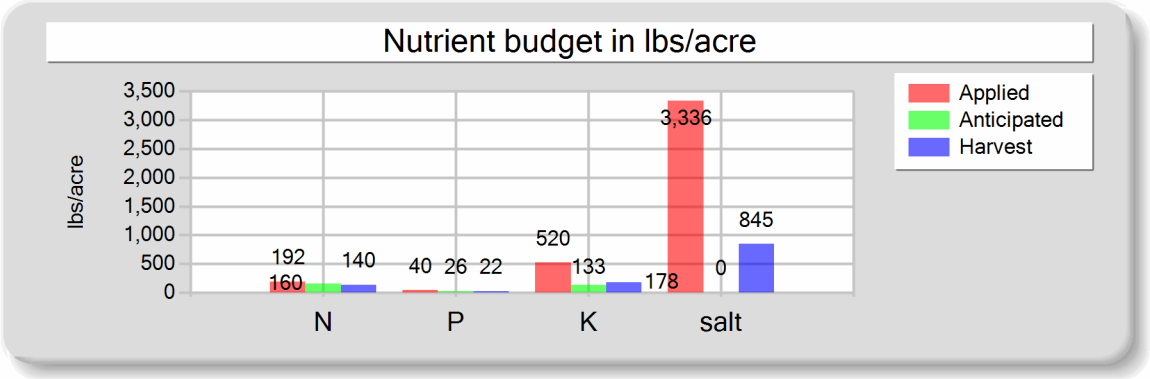
Fresh water applied	
14,322,424.00	*gallons*
527.45	*acre-inches*
37.67	*inches/acre*

Process wastewater applied	
2,322,000.00	*gallons*
85.51	*acre-inches*
6.11	*inches/acre*

Total harvests for the crop	
1	*harvests*

11 - 11/10/2022: Oats, silage-soft dough

Field name: 11 Crop: Oats, silage-soft dough Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	184.53	39.77	520.26	2,790.37
Fresh water	0.00	0.00	0.00	545.55
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	191.53	39.77	520.26	3,335.92
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	139.62	22.26	178.07	845.07
Nutrient balance	51.91	17.52	342.19	2,490.85
Applied to removed ratio	1.37	1.79	2.92	3.95

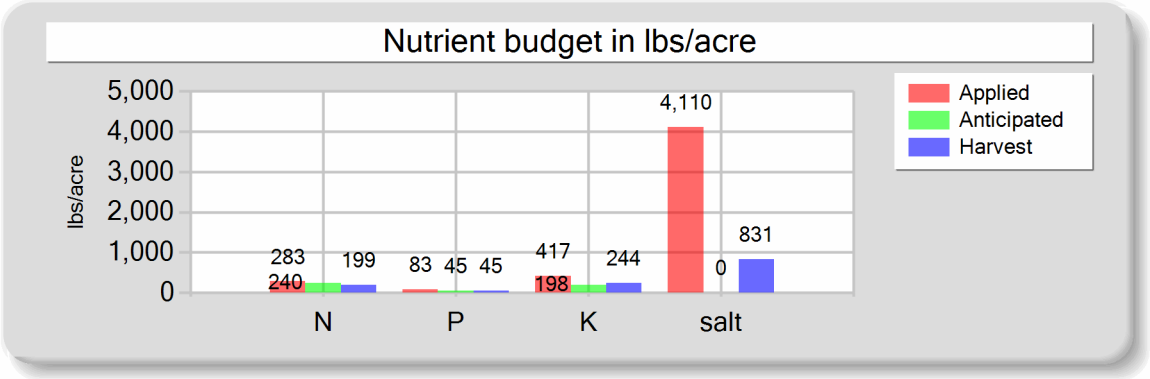
Fresh water applied	
5,169,139.00	*gallons*
190.36	*acre-inches*
11.20	*inches/acre*

Process wastewater applied	
1,440,000.00	*gallons*
53.03	*acre-inches*
3.12	*inches/acre*

Total harvests for the crop	
1	*harvests*

11 - 06/08/2023: Corn, silage

Field name: 11 Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	275.61	83.44	416.77	2,189.89
Fresh water	0.00	0.00	0.00	1,920.13
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	282.61	83.44	416.77	4,110.02
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	198.78	45.18	243.95	831.07
Nutrient balance	83.83	38.26	172.82	3,278.95
Applied to removed ratio	1.42	1.85	1.71	4.95

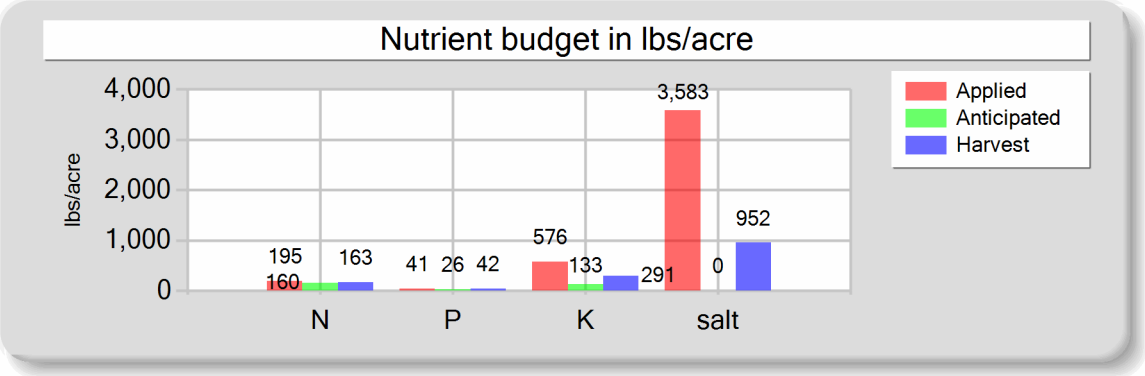
Fresh water applied	
18,193,452.00	*gallons*
670.00	*acre-inches*
39.41	*inches/acre*

Process wastewater applied	
2,736,000.00	*gallons*
100.76	*acre-inches*
5.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

15 - 11/10/2022: Oats, silage-soft dough

Field name: 15 Crop: Oats, silage-soft dough Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	188.36	40.64	575.75	3,027.23
Fresh water	0.00	0.00	0.00	555.83
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	195.36	40.64	575.75	3,583.06
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	163.20	41.60	291.20	951.81
Nutrient balance	32.16	-0.96	284.55	2,631.26
Applied to removed ratio	1.20	0.98	1.98	3.76

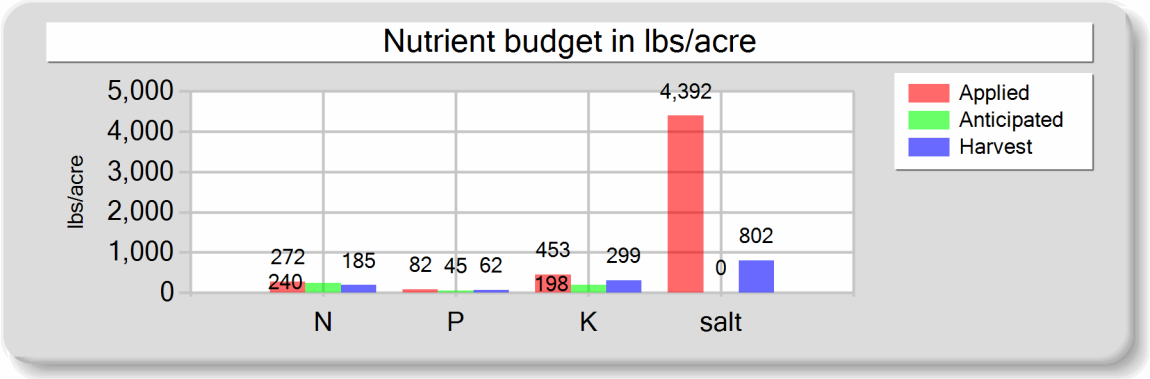
Fresh water applied	
929,394.00	*gallons*
34.23	*acre-inches*
11.41	*inches/acre*

Process wastewater applied	
270,000.00	*gallons*
9.94	*acre-inches*
3.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

15 - 06/08/2023: Corn, silage

Field name: 15 Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	264.97	81.56	452.88	2,496.49
Fresh water	0.00	0.00	0.00	1,895.88
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	271.97	81.56	452.88	4,392.37
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	184.80	61.60	299.20	802.03
Nutrient balance	87.17	19.96	153.68	3,590.33
Applied to removed ratio	1.47	1.32	1.51	5.48

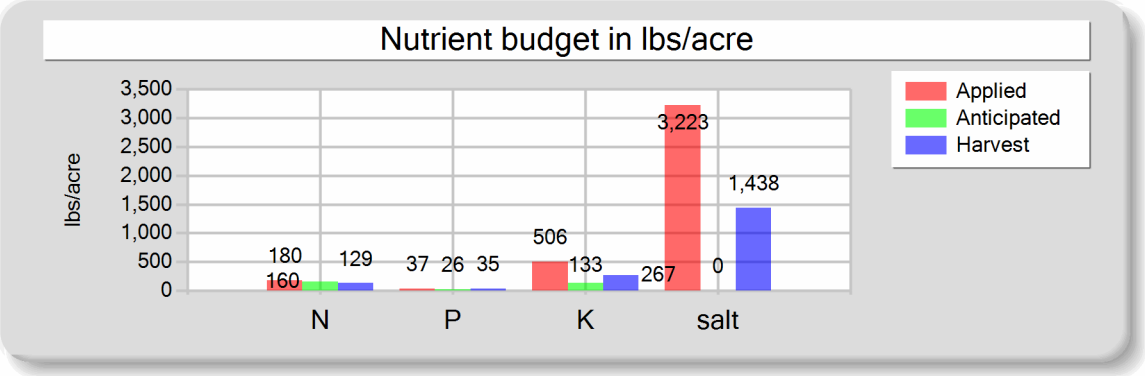
Fresh water applied	
3,170,051.00	*gallons*
116.74	*acre-inches*
38.91	*inches/acre*

Process wastewater applied	
486,000.00	*gallons*
17.90	*acre-inches*
5.97	*inches/acre*

Total harvests for the crop	
1	*harvests*

18 - 11/10/2022: Oats, silage-soft dough

Field name: 18 Crop: Oats, silage-soft dough Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	172.53	37.16	505.92	2,685.75
Fresh water	0.00	0.00	0.00	537.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	179.53	37.16	505.92	3,223.06
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	128.90	34.58	267.24	1,438.00
Nutrient balance	50.63	2.58	238.68	1,785.05
Applied to removed ratio	1.39	1.07	1.89	2.24

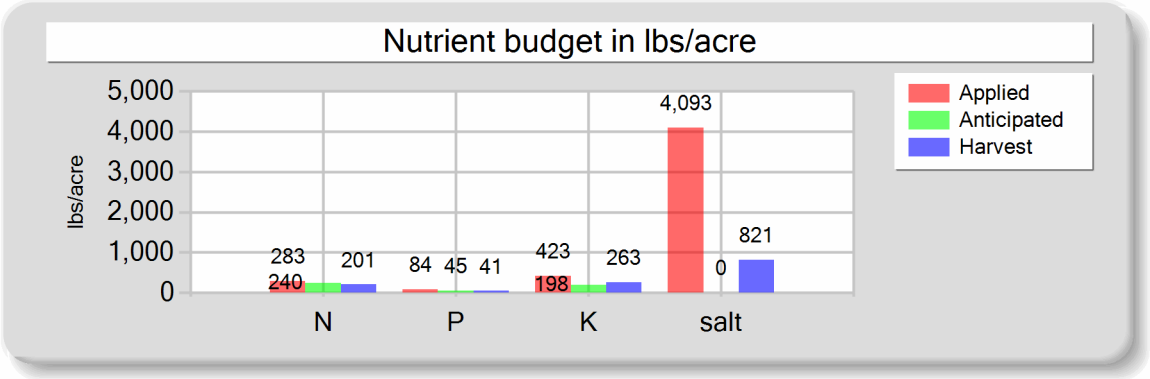
Fresh water applied	
7,486,750.00	*gallons*
275.71	*acre-inches*
11.03	*inches/acre*

Process wastewater applied	
2,016,000.00	*gallons*
74.24	*acre-inches*
2.97	*inches/acre*

Total harvests for the crop	
1	*harvests*

18 - 06/08/2023: Corn, silage

Field name: 18 Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	276.24	83.76	422.69	2,232.12
Fresh water	0.00	0.00	0.00	1,861.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	283.24	83.76	422.69	4,093.39
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	201.24	41.28	263.16	821.06
Nutrient balance	82.00	42.48	159.53	3,272.33
Applied to removed ratio	1.41	2.03	1.61	4.99

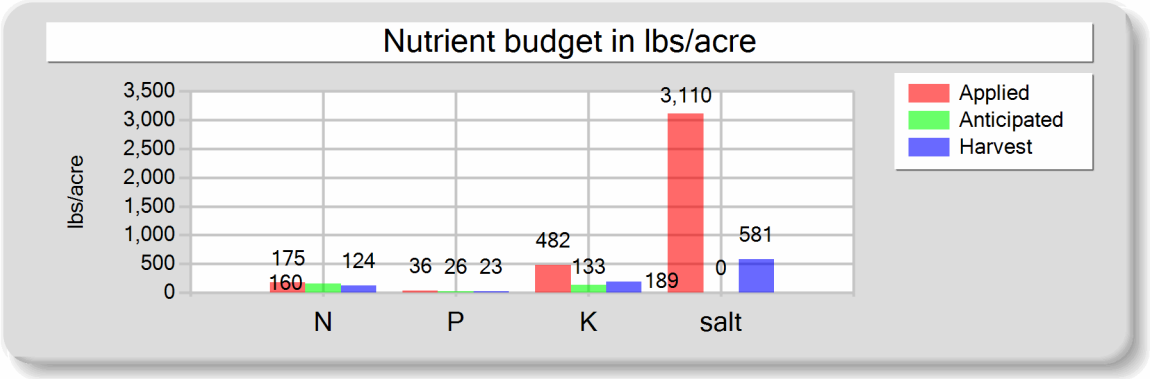
Fresh water applied	
25,934,921.00	*gallons*
955.09	*acre-inches*
38.20	*inches/acre*

Process wastewater applied	
4,050,000.00	*gallons*
149.15	*acre-inches*
5.97	*inches/acre*

Total harvests for the crop	
1	*harvests*

19 - 11/10/2022: Oats, silage-soft dough

Field name: 19 Crop: Oats, silage-soft dough Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	168.29	35.75	481.80	2,558.08
Fresh water	0.00	0.00	0.00	551.87
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	175.29	35.75	481.80	3,109.95
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	123.75	22.50	189.00	580.85
Nutrient balance	51.54	13.25	292.80	2,529.10
Applied to removed ratio	1.42	1.59	2.55	5.35

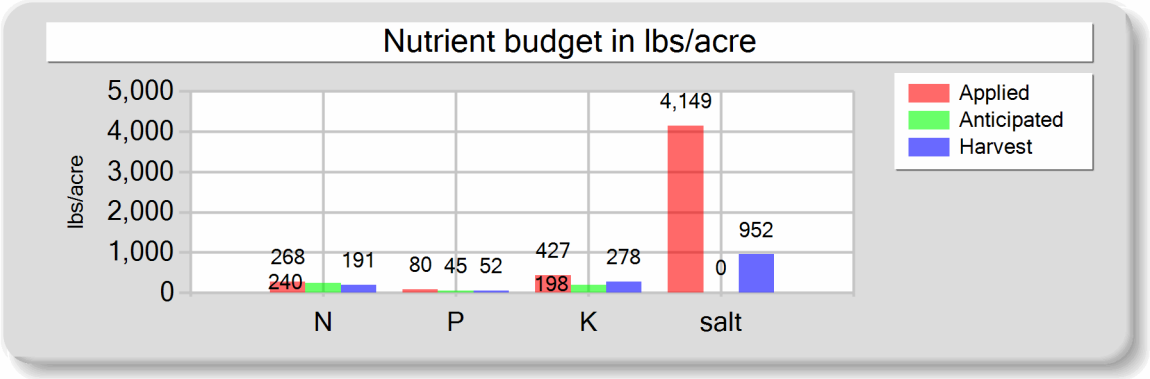
Fresh water applied	
2,460,736.00	*gallons*
90.62	*acre-inches*
11.33	*inches/acre*

Process wastewater applied	
612,000.00	*gallons*
22.54	*acre-inches*
2.82	*inches/acre*

Total harvests for the crop	
1	*harvests*

19 - 06/08/2023: Corn, silage

Field name: 19 Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	261.47	79.97	426.72	2,312.26
Fresh water	0.00	0.00	0.00	1,837.10
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	268.47	79.97	426.72	4,149.36
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	190.55	51.50	278.10	951.72
Nutrient balance	77.92	28.47	148.62	3,197.64
Applied to removed ratio	1.41	1.55	1.53	4.36

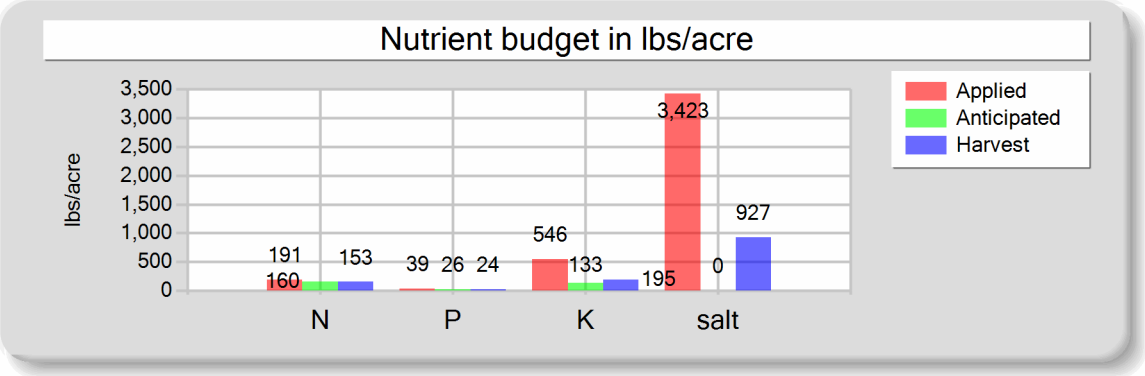
Fresh water applied	
8,191,393.00	*gallons*
301.66	*acre-inches*
37.71	*inches/acre*

Process wastewater applied	
1,256,400.00	*gallons*
46.27	*acre-inches*
5.78	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 11/10/2022: Oats, silage-soft dough

Field name: 5 Crop: Oats, silage-soft dough Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	184.23	38.92	545.88	2,866.91
Fresh water	0.00	0.00	0.00	556.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	191.23	38.92	545.88	3,423.18
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	153.18	24.42	195.36	927.12
Nutrient balance	38.05	14.50	350.52	2,496.07
Applied to removed ratio	1.25	1.59	2.79	3.69

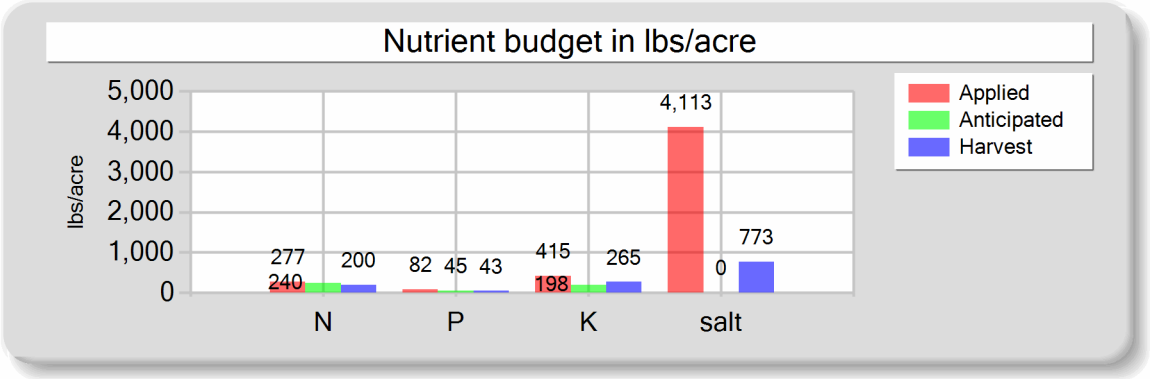
Fresh water applied	
3,100,442.00	*gallons*
114.18	*acre-inches*
11.42	*inches/acre*

Process wastewater applied	
846,000.00	*gallons*
31.16	*acre-inches*
3.12	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 06/08/2023: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	269.60	81.80	414.64	2,194.27
Fresh water	0.00	0.00	0.00	1,918.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	276.60	81.80	414.64	4,112.56
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	199.80	43.20	264.60	772.74
Nutrient balance	76.80	38.60	150.04	3,339.82
Applied to removed ratio	1.38	1.89	1.57	5.32

Fresh water applied	
10,691,805.00	*gallons*
393.74	*acre-inches*
39.37	*inches/acre*

Process wastewater applied	
1,584,000.00	*gallons*
58.33	*acre-inches*
5.83	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Spring Corral Solids

Sample and source description: Spring Corral Solids

Sample date: 05/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 31.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	3,100.00	10,600.00							40.50
DL	100.00	100.00	100.00							0.01

Spring Separated Solids

Sample and source description: Spring Separated Solids

Sample date: 05/01/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,600.00	1,600.00	3,300.00							18.50
DL	100.00	100.00	100.00							0.01

Fall Corral Solids

Sample and source description: Fall Corral Solids

Sample date: 10/23/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,800.00	5,000.00	18,000.00							81.80
DL	100.00	100.00	100.00							0.01

Fall Separated Solids

Sample and source description: Fall Separated Solids

Sample date: 10/23/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,500.00	5,700.00	4,600.00	17,400.00	4,500.00	1,600.00	6,700.00	1,080.00		32.60
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

B. PROCESS WASTEWATER ANALYSES

2022 WWSQ4

Sample and source description: 2022 WWSQ4

Sample date: 12/13/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	347.00	190.00	0.00	0.00	64.90	791.00								4,660.00	4,200
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ1

Sample and source description: WWSQ1

Sample date: 03/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	263.00	168.00	0.00	0.00	53.00	851.00								5,180.00	4,300
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ2

Sample and source description: WWSQ2

Sample date: 06/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	146.00	106.00	0.00	0.00	50.60	469.00								2,830.00	3,020
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ3

Sample and source description: WWSQ3

Sample date: 09/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	221.00	120.00	0.00	0.00	65.20	268.00	87.40	55.60	73.70	1,080.00	0.00	81.10	132.00	2,640.00	1,260
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

WWSQ4

Sample and source description: WWSQ4

Sample date: 12/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	829.00	496.00	0.00	0.00	76.20	1,170.00								9,780.00	10,800
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal

Canal

T.I.D Canal

Sample description: T.I.D Canal

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	215
DL	0.50	0.50	0.50								1.00	10

Irrigation Well

Irrigation Well

Sample description: Irrigation Well

Sample date: 09/19/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	26.60	0.00	23.80								721.00	400
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

10 - 11/10/2022: Oats, silage-soft dough

10 - 11/10/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,900.00	1,100.00	8,800.00		13.30
DL	100.00	100.00	100.00		0.01

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,000.00	1,100.00	8,800.00		8.15
DL	100.00	100.00	100.00		0.01

11 - 11/10/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,200.00	1,400.00	11,500.00		8.85
DL	100.00	100.00	100.00		0.01

11 - 06/08/2023: Corn, silage

11 - 06/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,400.00	1,000.00	5,400.00		6.30
DL	100.00	100.00	100.00		0.01

15 - 11/10/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,100.00	1,300.00	9,100.00		10.40
DL	100.00	100.00	100.00		0.01

15 - 06/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,700.00	800.00	4,800.00		5.50
DL	100.00	100.00	100.00		0.01

15 - 06/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	1,400.00	6,800.00		6.20
DL	100.00	100.00	100.00		0.01

18 - 11/10/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 75.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	1,100.00	8,500.00		18.90
DL	100.00	100.00	100.00		0.01

18 - 06/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	800.00	5,100.00		5.20
DL	100.00	100.00	100.00		0.01

19 - 11/10/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,500.00	1,000.00	8,400.00		9.09
DL	100.00	100.00	100.00		0.01

19 - 06/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,700.00	1,000.00	5,400.00		6.60
DL	100.00	100.00	100.00		0.01

5 - 11/10/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 75.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,700.00	1,000.00	7,700.00		14.50
DL	100.00	100.00	100.00		0.01

5 - 06/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,700.00	800.00	4,900.00		4.50
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

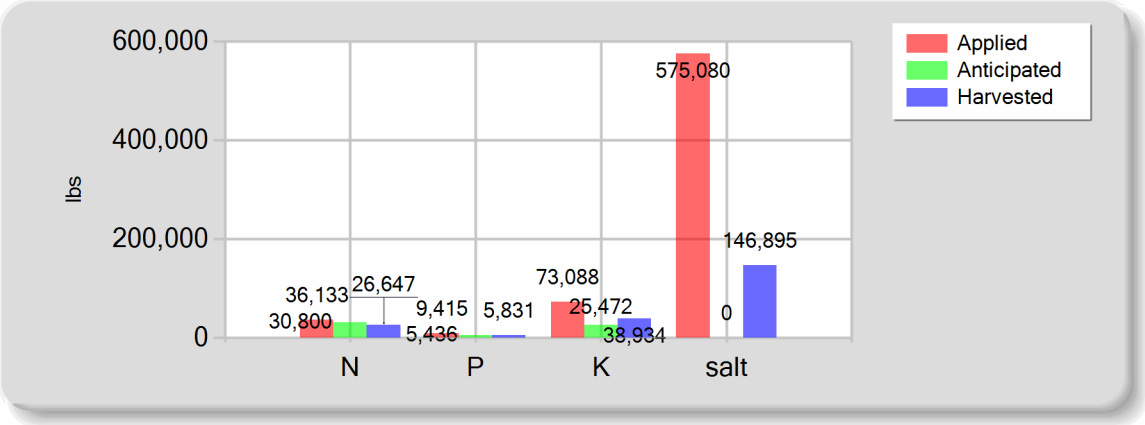
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

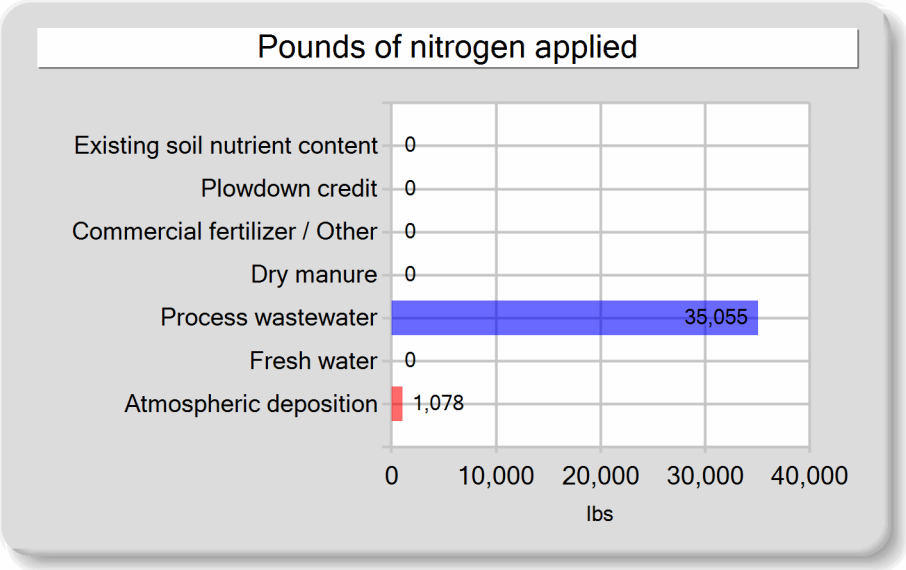
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

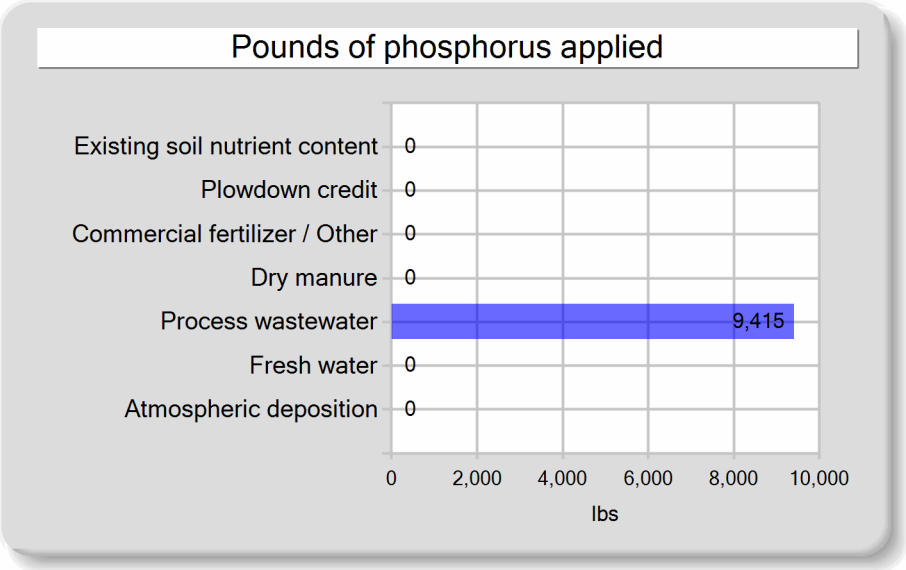
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	35,054.93	9,414.72	73,087.59	388,477.31
Fresh water	0.00	0.00	0.00	186,602.54
Atmospheric deposition	1,078.00	0.00	0.00	0.00
Total nutrients applied	36,132.93	9,414.72	73,087.59	575,079.85
Anticipated crop nutrient removal	30,800.00	5,436.20	25,471.60	0.00
Actual crop nutrient removal	26,646.80	5,831.00	38,934.00	146,895.10
Nutrient balance	9,486.13	3,583.72	34,153.59	428,184.75
Applied to removed ratio	1.36	1.61	1.88	3.91

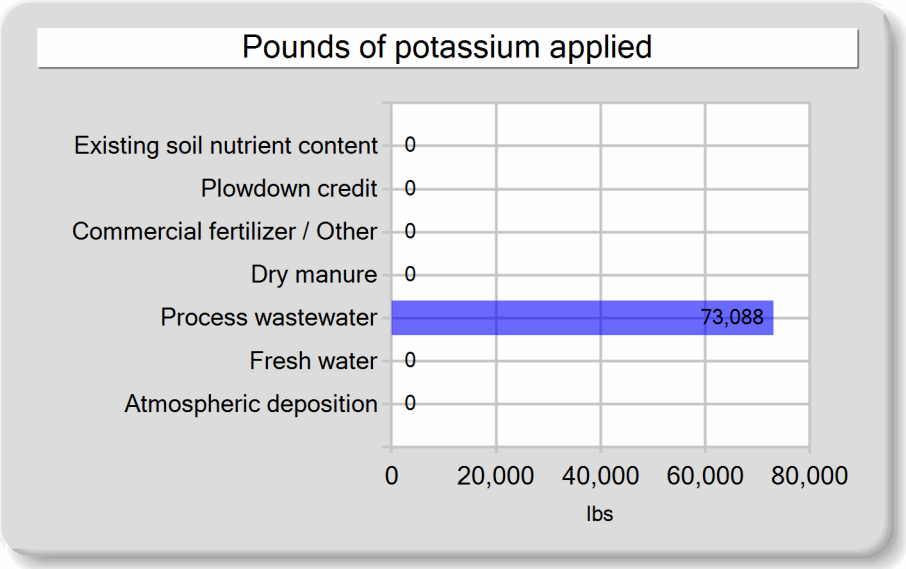
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

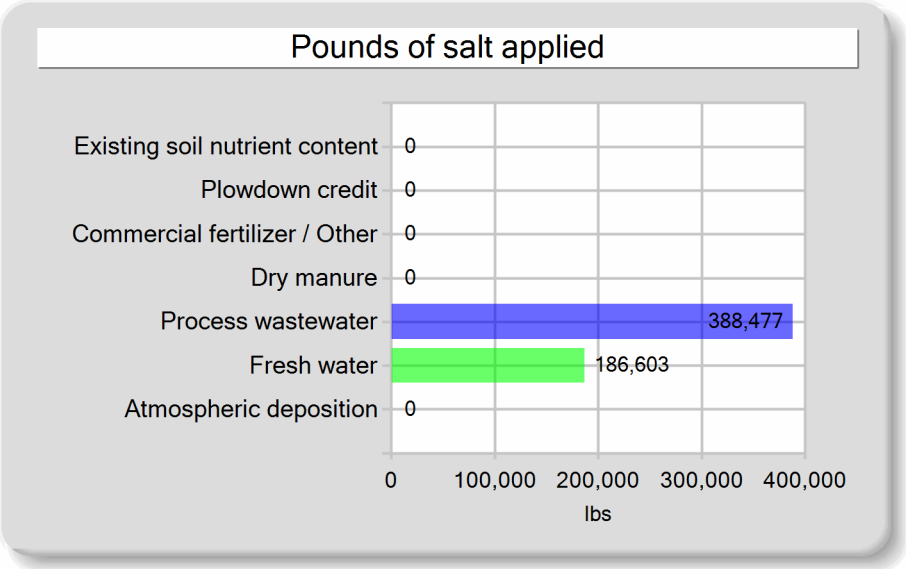


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Almond 1 are non bearing trees. No production in 2023

Fields 11, 15, 18, 19 corn silage had lower than anticipated harvested tons (22-25.8 tons per acre harvest vs 30 tons per acre anticipated) caused these fields to be slightly above the 1.4 threshold.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Lloyd Fagundes Lloyd Fagundes (Jun 25, 2024 12:52 PDT)	Larry A. Matos Larry A. Matos (Jun 27, 2024 09:23 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
/Fagundes, Fagundes, Fagundes GP Forebay Farms LLC	California Dairy Farms
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 25, 2024	Jun 27, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
CDF Howard
P.O. Box 1090
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: Hilmar
RECEIVED DATE: 9/19/23 3:00 PM
SUBMITTED BY: KJ
ANALYZED DATE: 9/21/23

Process Waste Water - H3 & H5 (H8)
SOURCE: Lagoon Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units	Lbs/10000 Gal as Rec'd
WWSQ3	T3262039A	Electrical Conductivity (EC)	EPA 120.1 and	2.64	mmhos/cm	N/A
		Bicarbonate Alkalinity (ALK-B)	SM 2320B-2011	1080	mg/L	90.1
		Carbonate Alkalinity (Carb)	SM 2320B-2011	< 2.00	mg/L	0.000
		Total Phosphorus (TP)	EPA 200.7	65.2	mg/L	5.44
		Calcium (Ca)	EPA 200.7	87.4	mg/L	7.29
		Sodium (Na)	EPA 200.7	73.7	mg/L	6.15
		Total Dissolved Solids (TDS)	SM 2540 C-1997	126000	mg/L	10500
		Sulfate (SO4)	EPA 375.2	81.1	mg/L	6.77
		Chloride (Cl)	SM 4500-Cl D-2011	132	mg/L	11.0
		Magnesium (Mg)	EPA 200.7	55.6	mg/L	4.64
		Total Kjeldahl Nitrogen (TKN)	SM 4500-NH3 C-	221	mg/L	18.4
		Soluble Salts (SALT-SOL)	Calculation	1690	mg/L	141
		Total Potassium (TK)	EPA 200.7	268	mg/L	22.4
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 C-	120	mg/L	10.0

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	97.4			120 (T3262039A)	120 (null)	0.000
Calcium (Ca)	93.6	93.6 (null)	92.4 (null)			1.30
Chloride (Cl)	94.8			68.0 (T3264009A)	65.1 (null)	4.22
Magnesium (Mg)	98.2	85.0 (null)	84.6 (null)			0.453
Sodium (Na)	90.7	110 (null)	84.3 (null)			26.3
Sulfate (SO4)	99.5			91.4 (T3264009A)	45.7 (null)	66.6
Total Dissolved Solids (TDS)	94.0			1600 (T3263024A)	1660 (Dup)	3.68
Total Kjeldahl Nitrogen (TKN)	99.1			221 (T3262039A)	227 (null)	2.50
Total Phosphorus (TP)	90.8	95.6 (null)	95.6 (null)			0.000
Total Potassium (TK)	97.1	96.8 (null)	95.8 (null)			1.01

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ELAP Certificate No.: 2714

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Compliance Analysis Report

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CDF Howard
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Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: Hilmar
RECEIVED DATE: 9/19/23 3:00 PM
SUBMITTED BY: KJ
ANALYZED DATE: 9/21/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 21.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Domestic Wells 1	T3262039B	Electrical Conductivity (EC)	EPA 120.1 and	0.464	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	297	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	96.6	94.0 (T3262039B)	98.8 (T3262039B)	4.97
Electrical Conductivity (EC)	99.3			0.475
Nitrate Nitrogen (NO3-N)	95.5	93.5 (T3262023A)	92.0 (T3262023A)	1.62

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Compliance Analysis Report

Customer
CDF Howard
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Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: Hilmar
RECEIVED DATE: 9/19/23 3:00 PM
SUBMITTED BY: KJ
ANALYZED DATE: 9/21/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 21.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Domestic Wells 2	T3262039C	Electrical Conductivity (EC)	EPA 120.1 and	0.768	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	492	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	96.6	94.0 (T3262039B)	98.8 (T3262039B)	4.97
Electrical Conductivity (EC)	99.3			0.475
Nitrate Nitrogen (NO3-N)	95.5	93.5 (T3262023A)	92.0 (T3262023A)	1.62

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Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: Hilmar
RECEIVED DATE: 9/19/23 3:00 PM
SUBMITTED BY: KJ
ANALYZED DATE: 9/21/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 21.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Domestic Wells 4	T3262039D	Electrical Conductivity (EC)	EPA 120.1 and	0.771	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	493	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	96.6	94.0 (T3262039B)	98.8 (T3262039B)	4.97
Electrical Conductivity (EC)	99.3			0.475
Nitrate Nitrogen (NO3-N)	95.5	93.5 (T3262023A)	92.0 (T3262023A)	1.62

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Compliance Analysis Report

Customer
CDF Howard
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Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: Hilmar
RECEIVED DATE: 9/19/23 3:00 PM
SUBMITTED BY: KJ
ANALYZED DATE: 9/21/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 21.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Domestic Wells 6	T3262039E	Electrical Conductivity (EC)	EPA 120.1 and	0.769	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	492	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	96.6	94.0 (T3262039B)	98.8 (T3262039B)	4.97
Electrical Conductivity (EC)	99.3			0.475
Nitrate Nitrogen (NO3-N)	95.5	93.5 (T3262023A)	92.0 (T3262023A)	1.62

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2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: Hilmar
RECEIVED DATE: 9/19/23 3:00 PM
SUBMITTED BY: KJ
ANALYZED DATE: 9/21/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 21.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Irrigation 1	T3262039F	Electrical Conductivity (EC)	EPA 120.1 and	0.721	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	23.8	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	400	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	26.6	mg/L
		Soluble Salts (SALT-SOL)	Calculation	461	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	96.6	94.0 (T3262039B)	98.8 (T3262039B)			4.97
Electrical Conductivity (EC)	99.3			0.464 (T3262039B)	0.462 (Dup)	0.475
Nitrate Nitrogen (NO3-N)	95.5	93.5 (T3262023A)	92.0 (T3262023A)			1.62
Total Dissolved Solids (TDS)	100			1500 (T3268021J)	1440 (Dup)	4.43
Total Nitrogen (TN)	97.4			2.80 (T3261032A)	2.80 (null)	0.000

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: CDF-HILMGR
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: F & R Ag Services

Sampled By: K.Jones
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	9/19	11:30	Wastewater	WWSQ3	24.						H8
2	9/19	8:00	Domestic wells	DW1	21.8°		✓				
3		8:15		DW2	21.6°		✓				
4		8:30		DW4	21.0°		✓				
5		8:45		DW6	21.2°		✓				
6		9:00	Irrigation well	IW1	21.6°						H7
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st	Kristen Jones	F & R Ag Services				
2nd		DENELE ANALYTICAL, INC.	9-19-23 3:00			T3262039

Required to be submitted to any agency? (YES / NO)

2023 Annual Report-CDF Hilmar

Final Audit Report 2024-06-27

Created:	2024-06-19
By:	Kristen Jones (kjones@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAA4uEbZvlghQe5zVR-3w-xvBv8VPKjv_1i

"2023 Annual Report-CDF Hilmar" History

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